mini elements

_travelers guide
_for zero gravity botany

3	5	7
exciters from elements	menu pages	octave steps, locked

[poly] press · **[poly]** hold · › then · noon = 12 o'clock · cw / ccw = clockwise / counter

elements' exciter, borrowed into rings · a menu behind the two buttons · free alt firmware for rings

installs like any rings update · long press = 3 s

orange = hold

01 buttons

rng_01 · stock gestures kept

[poly]	voices 1 · 2 · 4 · in the menu: next page
[model]	resonator · in the menu: next value
[poly] 3 s	3 voices, odd / even rhythm (as stock)
[model] 3 s	the hidden twin of this model (as stock)
tap **[poly] + [model]**	frequency lock on / off (02)
[poly] › [model]	3 s: **options menu** (**[poly]** went down first)
[model] › [poly]	3 s: **calibration** (**[model]** went down first)
in the menu: **[poly] + [model]**	saves and leaves (LEDs flicker red / green)
power on + [model]	colour-blind mode, as stock

02 frequency lock

rng_02 · tap both

tap **[poly] + [model]**	all LEDs flash: red = will lock · green = will unlock · it flips when you let go
locked	**[frequency]** knob stops tuning · v/oct, frequency cv and its attenuverter still work
locked · rings modes	**[frequency]** = octave switch: 7 steps, –3 to +3 octaves
locked · mini-elements	**[frequency]** = exciter **timbre**
remembered	the lock and its tuning survive power off

06 hold on strum

rng_06 · menu page 3

off	stock: the knobs act live
on	each strum samples structure, brightness, damping and position (and most exciter settings) · frequency stays live
with voices	only the new voice takes the new values, so every voice rings with its own settings
a strum is	a **[strum]** trigger, or a v/oct note change when strum is empty

09 good to know

rng_09

32 kHz inside	down from 48 kHz, as on elements: nothing above 16 kHz
no slew	the structure, brightness, damping and position cv are no longer smoothed, so fast cv bites harder

03 five modes

rng_03 · menu page 1

classic rings	stock rings, stereo
rings + waveform	mono rings on **[odd]** · **[even]** = square, impulse train or saw at the resonator pitch: tune, sync, feedback
mini-elements	elements' exciter plays the resonator inside · stereo
mini-elements + exciter	resonator on **[odd]** · raw exciter on **[even]**: envelope it, patch it back into **[in]**
easter egg	disastrous peace · the knob handshake is gone, the menu is the only way in

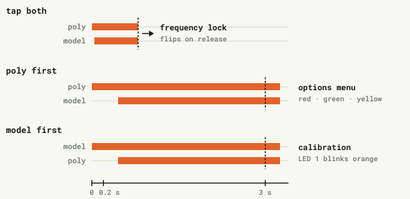
07 outputs

rng_07 · odd / even by mode

classic · mini-elements	stereo, as stock: split by partials or by voice
rings + waveform	**[odd]** = rings (mono) · **[even]** = the waveform
mini-el. + exciter	**[odd]** = resonator · **[even]** = raw exciter
in the two + modes	always patch **[even]**: unpatched, the hardware mixes it into **[odd]**

_fig.01 : three gestures

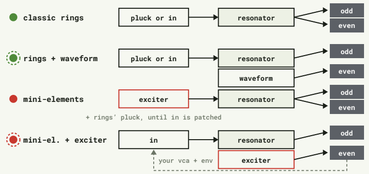
two buttons, held together



between 0.2 s and 3 s the lock waits for you to let go · past 3 s, whichever button went down first

_fig.06 : four ways out

menu page 1, drawn



04 the exciters

rng_04 · menu page 2

bow

continuous · menu page 2

timbre	bow colour, dark → bright
position	still the strike point
strum	not needed: it sounds all the time, more oscillator than resonator

blow

continuous · through elements' tube

timbre	pitch of the breath grains
position	**flow**: where the breath recording restarts
damping	also damps the tube

strike

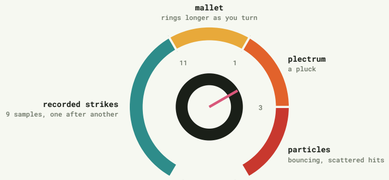
needs triggers or gates on strum

timbre	how bright the hit is
position	**mallet**: recorded strikes › mallet › plectrum › particles (fig.04)
strum	each trigger or gate fires the exciter

timbre = **[frequency]** once locked, or the frequency cv (menu page 5) · one exciter at a time, no envelope knob · rings' own pluck still plays: a dummy cable in **[in]** silences it

_fig.04 : the strike dial

position = mallet



strike mode only · timbre (frequency, once locked) = how bright the hit is

08 install · calibrate

rng_08

install

free · a .wav, like any rings update

1. download rings_mini_elements.wav · lines thread
2. power on holding **[poly]** · play the file into **[in]** · **[frequency]** = gain
3. back to stock: the rings 1.1 .wav, same way

retune the v/oct

hold **[model]** first, or you land in the menu

1. unplug all cv · keyboard into **[v/oct]** · a cable in the **[frequency]** cv jack, other end loose
2. hold **[model]**, add **[poly]**, keep both 3 s › LED 1 blinks orange
3. send 1 v · press **[poly]** › send 3 v · press **[poly]** › done

no calibration on copies downloaded before 2022.02.27: download it again

11 who made it

rng_11 · credits

firmware	lylepmills, on émilie gillet's open rings + elements code
chord tables	jon butler · joe mcmullen

05 chord tables

rng_05 · menu page 4

classic

the stock tables

jon butler

after his plaits chords

chords	reorganised, with a few more
3–4 voices	western chords plays the first 16 intervals instead, unison to major 10th

joe mcmullen

five tables in one

easter egg	his plaits chord table, revoiced for each voice count
1 voice	15 scales (fig.05): a tuned reverb for whatever you feed into **[in]**
2 voices	21 chords of one key, descending in thirds: c am f dm b° g em …
3 voices	the first 16 harmonics of the overtone series
4 voices	23 just intervals after the 22 shrutis, 1/1 to 2/1

used by the easter egg and by western chords (the sympathetic strings twin) · **[frequency]** sets the key, **[structure]** picks the chord

_fig.05 : the scale dial

joe mcmullen · 1 voice



1	hungarian minor	9	major triad
2	lydian minor	10	pentatonic major
3	melodic minor	11	diatonic major
4	harmonic minor	12	harmonic major
5	diatonic minor	13	melodic major
6	pentatonic minor	14	phrygian major
7	minor triad	15	byzantine major
8	perfect fifths		

minor side ccw · major side cw
set [frequency] to the key you play in

10 first patches

rng_10

bowed drone
mini-elements · bow · lock it
[frequency] now colours the bow · no strum needed, it never stops

breath pad
mini-elements · blow
[position] = flow · damping long

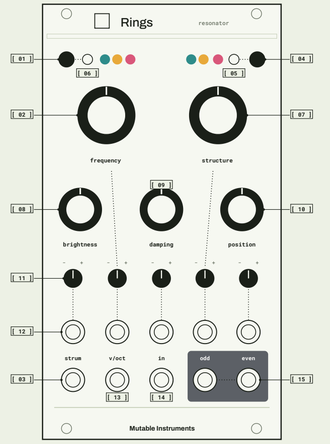
a new body every hit
strike · hold on strum on
random cv into structure, triggers into **[strum]**: each voice keeps its own body

shape the exciter
mini-elements + exciter
[even] › vca with an envelope › **[in]**

tuned reverb
[model] 3 s on sympathetic · joe table · 1 voice
any voice into **[in]** · **[frequency]** = the key

_fig.03 : front panel

what each part does here



[01] _poly : voices · menu page
[02] _frequency : timbre or octaves, once locked
[03] _strum : strike needs it
[04] _model : resonator · menu value
[05] _model LED : menu value
[06] _poly LED : menu page

[07] _structure : body · chord · scale
[08] _brightness : material
[09] _damping : decay · tube damping
[10] _position : strike point · flow · mallet
[11] _atts : one per knob
[12] _cv ins : frequency cv can drive timbre
[13] _v/oct : pitch · strums
[14] _in : dummy cable = no pluck
[15] _odd / even : see 07

_fig.02 : the options menu

left LED = page · right LED = value

mode page 1	classic rings · rings + waveform · mini-elements · mini-elements + exciter · easter egg
waveform / exciter page 2	square · bow · impulse train · blow · saw · strike
hold on strum page 3	off · on
chord table page 4	classic · jon butler · joe mcmullen
frequency cv page 5	pitch, as stock · exciter timbre

[poly] › **[model]** 3 s = in · **[poly]** = next page · **[model]** = next value
[poly] + **[model]** = save + out

= blinking · in the menu the poly LED flashes every 10 s as a reminder